Notes: Loutskina and Strahan (JF, 2009)

Securitization and the Declining Impact of Bank Finance on Loan Supply: Evidence from Mortgage Originations

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1 Big picture

- **Question.** Does securitization weaken the link between a bank's own financial condition and its supply of credit?
- **Core idea.** Banks can fund very liquid loans either with deposits or by quickly selling or securitizing them. Illiquid loans must be held and funded on balance sheet. If so, bank liquidity and deposit costs should matter much more for illiquid loans than for liquid ones.
- **Empirical setting.** The paper studies residential mortgages. Conforming, or "nonjumbo," mortgages can be purchased by Fannie Mae and Freddie Mac, while jumbo mortgages cannot. This charter rule creates a sharp difference in secondary-market liquidity around the jumbo cutoff.
- **Identification strategy.** Compare each bank's lending behavior in the nonjumbo segment to its behavior in the jumbo segment within the same year. The maintained assumption is that unobserved demand conditions affect both segments similarly enough that differencing can sweep them out.
- **Main volume result.** Banks with more liquid balance sheets originate relatively more jumbo mortgages. In the benchmark volume regression, moving liquid assets from the 25th to the 75th percentile increases jumbo originations relative to nonjumbos by about 2.8% of assets in the full sample and about 1.7% of assets in the sample without refinancings.
- **Main funding-cost result.** Banks with lower deposit costs also originate relatively more jumbo mortgages. An increase in deposit costs from the 25th to the 75th percentile shifts originations toward nonjumbos by a little more than 0.8% of assets in the full sample and about 0.5% of assets in the no-refi sample.
- **Main approval result.** Banks with more liquidity and cheaper deposits are more willing to approve jumbo mortgages than otherwise similar banks, but those variables have little or no effect on nonjumbo approvals.
- **Sorting result.** Approval behavior explains only part of the origination result. Financially stronger banks also receive relatively more jumbo applications, especially right around the cutoff, consistent with borrower sorting and loan-size adjustment.
- **Validation result.** The finance effects appear at the jumbo cutoff, not at arbitrary places such as 75% or 150% of the cutoff. A separate homogeneous-sample test using loans that switch classification when the annual cutoff rises delivers the same pattern.
- **Broader takeaway.** Securitization changes banking from "originate and hold" toward "originate to distribute." As loans become more liquid, shocks to deposit supply or bank balance-sheet liquidity matter less for credit supply. That weakens the traditional bank lending channel and dampens the local consequences of bank-specific funding shocks.

2 Data and measurement (key objects)

2.1 Mortgage application data

The core data come from HMDA and cover mortgage applications from 1992 to 2004. The paper begins with roughly 250 million applications and then narrows the sample to commercial banks.

The main sample restrictions are:

- drop savings institutions, mortgage bankers, credit unions, and other nonbank lenders;
- drop mortgages backed by the FHA, VA, and other government programs;
- drop applications with missing loan size, location, or approval status;
- merge the remaining applications to the bank's Call Report from the fourth quarter of the previous year.

After these steps, the **full sample** contains about 62.6 million mortgage applications. The paper also constructs a tighter **filtered sample** that drops refinancing loans and keeps only applications between 50% and 250% of the jumbo cutoff. This leaves about 6.4 million applications.

Interpretation. The full sample gives maximum power. The filtered sample sacrifices size in exchange for a much more homogeneous pool of loans around the policy-relevant cutoff.

2.2 Why the jumbo cutoff is the key measurement object

The GSEs can buy conforming mortgages but not jumbo mortgages. As a result:

- **nonjumbos** are relatively liquid because they can be sold or securitized more easily;
- **jumbos** are relatively illiquid because they cannot be sold to the GSEs and are more likely to remain on balance sheet or be financed in thinner private markets.

During the sample period, the jumbo cutoff rises from about $202,000 in 1992 to about $334,000 in 2004.

Interpretation. The paper does not need all nonjumbos to be perfectly liquid and all jumbos to be perfectly illiquid. It only needs liquidity to fall discretely at the cutoff. If some nonjumbos are also hard to sell, that works against finding the predicted effect.

2.3 Figure 1: what the raw data already show

Figure 1 plots the frequency of applications and the acceptance rate by loan size relative to the jumbo cutoff.

Two facts stand out:

- there is a sharp spike in application volume just *below* the cutoff;
- there is also a sharp spike in approval rates just *below* the cutoff.

Interpretation. The discontinuity confirms that the cutoff matters economically. At the same time, it warns us that borrowers and lenders may actively manage loan size near the boundary. This is why the paper later studies approval rates, application sorting, and a near-cutoff homogeneous sample.

2.4 Bank financial variables

The bank-side variables come from Call Reports.

The two central variables are:

- **Balance sheet liquidity** = cash plus marketable securities divided by assets.
- **Cost of deposits** = interest expense on deposits divided by total deposits.

Additional bank controls are the log of bank assets, an indicator for bank ownership by a holding company, capital divided by assets, and net income divided by assets.

Interpretation. Liquidity captures whether the bank has balance-sheet resources to hold illiquid loans. Deposit cost captures how expensive it is to fund loans that cannot be quickly sold.

2.5 Summary statistics

From Table I, the full application sample has an average acceptance rate of about 79.9%, average loan size of about $102,000, and average applicant income of about $79,000. In the filtered sample, the average acceptance rate rises to about 90.9%, average loan size rises to about $209,000, and the average loan-to-income ratio rises from 1.60 to 2.40.

From Table II, the median bank in the matched sample has:

- about $84.4 million in assets,
- liquid assets equal to about 27.8% of assets,
- deposit cost of about 3.3%,
- capital equal to about 8.5% of assets,
- nonjumbo originations equal to about 7.74% of assets,
- jumbo originations equal to about 1.38% of assets.

The median bank thus originates about 5.57% of assets more in nonjumbos than jumbos in the full sample.

Interpretation. Nonjumbos dominate in quantity because most mortgages fall below the conforming cutoff. The paper's argument is therefore about *relative sensitivity* to bank finance, not about jumbos becoming the larger market.

2.6 Loan-pool and market controls

HMDA does not contain everything an ideal credit-risk model would have. The paper therefore builds bank-year controls from the application pool itself. These include:

- loan-to-income ratio,
- log applicant income,
- share of properties in MSAs,

- median income in the property's census tract,
- shares of minority and female applicants,
- share of minority population in the property's tract.

Market controls include the log of average application size in the lender's market, the local unemployment rate, local income growth, and the share of the population over age 65, along with year and state fixed effects.

Interpretation. These controls are designed to absorb cross-market differences in house prices, local business conditions, demographics, and observable risk that might otherwise contaminate the funding-supply relationship.

3 Models and empirical specifications (all equations + interpretation)

3.1 Separate regressions for nonjumbo and jumbo originations

The paper starts from two reduced-form equations, one for the liquid segment and one for the illiquid segment:

$$VOL_{it}^{NJ} = \lambda_1^{NJ} Liq_{i,t-1} + \lambda_2^{NJ} DepCost_{i,t-1} + \beta_1^{NJ} Risk_{it}^{NJ} + \beta_2^{NJ} Controls_{i,t-1} + D_{it} + \varepsilon_{it}^{NJ}, \quad (1)$$

$$VOL_{it}^{J} = \lambda_1^{J} Liq_{i,t-1} + \lambda_2^{J} DepCost_{i,t-1} + \beta_1^{J} Risk_{it}^{J} + \beta_2^{J} Controls_{i,t-1} + D_{it} + \varepsilon_{it}^{J}. \quad (2)$$

Here, VOL_{it}^{NJ} and VOL_{it}^{J} are bank-year originations of approved nonjumbo and jumbo mortgages, respectively, each scaled by lagged assets. The term D_{it} represents demand-side forces common to both segments for bank i in year t.

Interpretation. These equations make the conceptual point clear: loan supply in each segment depends on bank funding conditions, borrower-pool characteristics, and market demand. The identification challenge is that D_{it} is not directly observed.

3.2 Benchmark difference specification

The main regression differences the two equations:

$$\begin{aligned} VOL_{it}^{NJ} - VOL_{it}^{J} = {} & (\lambda_1^{NJ} - \lambda_1^{J}) Liq_{i,t-1} + (\lambda_2^{NJ} - \lambda_2^{J}) DepCost_{i,t-1} \\ & + \left(\beta_1^{NJ} Risk_{it}^{NJ} - \beta_1^{J} Risk_{it}^{J}\right) + (\beta_2^{NJ} - \beta_2^{J}) Controls_{i,t-1} + u_{it}. \end{aligned} \quad (3)$$

The paper expects:

$$0 \leq \lambda_1^{NJ} < \lambda_1^{J}, \qquad 0 \geq \lambda_2^{NJ} > \lambda_2^{J}. \quad (4)$$

Because the dependent variable is nonjumbo−jumbo, the signs in the estimated difference regression should be:

- **negative on liquidity**: more bank liquidity supports jumbos more than nonjumbos;

- **positive on deposit cost**: higher funding costs hurt jumbos more than nonjumbos.

Interpretation. This is the paper's central estimating equation. It does not try to estimate the level effect of funding conditions on all lending. It estimates the *relative* effect on illiquid versus liquid mortgage supply.

3.3 Acceptance-rate specification

The next dependent variable is the difference in approval rates:

$$AR_{it}^{NJ} - AR_{it}^{J} = \delta_1 Liq_{i,t-1} + \delta_2 DepCost_{i,t-1} + X_{it}'\Pi + \eta_{it}. \tag{5}$$

Here, AR_{it}^{NJ} and AR_{it}^{J} are the fractions of applications approved in the two segments.

Interpretation. Approval rates are closer to the lender's own decision rule than raw origination volume. They also divide by the flow of applications, which helps purge demand effects. If bank funding conditions still matter here, the paper has stronger evidence of a supply mechanism.

3.4 Application-sorting specifications

To study sorting around the cutoff, the paper constructs two shares.

The **marginal nonjumbo share** is:

$$Share_{it}^{M} = \frac{N_{it}(0.95 \leq Loan/Cutoff < 1.00)}{N_{it}(0.95 \leq Loan/Cutoff < 1.05)}. \tag{6}$$

The **nonmarginal nonjumbo share** is:

$$Share_{it}^{NM} = \frac{N_{it}(0.50 \leq Loan/Cutoff < 0.95)}{N_{it}(0.50 \leq Loan/Cutoff < 0.95) + N_{it}(1.05 \leq Loan/Cutoff < 2.50)}. \tag{7}$$

These shares are then regressed on the same bank financial variables and controls.

Interpretation. If financially strong banks receive more jumbo applications, the origination effect is not coming only from approval behavior. It is also coming from where borrowers choose to apply and how they size their loans.

3.5 Away-from-cutoff bin interactions

To test whether the effect is really about the jumbo boundary rather than loan size in general, the paper creates four bins: 50–75%, 75–100%, 100–150%, and 150–250% of the cutoff.

A generic version of the specification is:

$$Y_{ibt} = \alpha_b + \sum_b \Big[\theta_b\big(Liq_{i,t-1} \times 1\{b\}\big) + \phi_b\big(DepCost_{i,t-1} \times 1\{b\}\big)\Big] + Z_{it}'\Gamma + \varepsilon_{ibt}, \tag{8}$$

where Y_{ibt} is either volume or acceptance rate for bank i, bin b, and year t.

Interpretation. The key question is not whether funding conditions matter somewhere in the size distribution. It is whether the *shift* in sensitivity occurs exactly where loan liquidity changes: at the jumbo cutoff.

3.6 Homogeneous-sample probit

The last design uses only loans that change classification in adjacent years when the conforming limit increases. The approval equation is a probit estimated at the individual-loan level:

$$\Pr(Approve_{ijt} = 1) = \Phi\big(\alpha + \beta_1 Jumbo_{ijt} + \beta_2 Liq_{i,t-1} + \beta_3 Jumbo_{ijt} \times Liq_{i,t-1} + \beta_4 DepCost_{i,t-1} + \beta_5 Jumbo_{ijt} \times DepCost_{i,t-1} + W'_{ijt}\Gamma\big). \tag{9}$$

The paper reports marginal effects rather than raw probit coefficients.

Interpretation. This is the cleanest homogeneity check in the paper. By looking only at loans whose jumbo status changes because the rule changes, the paper sharply limits concern that the results are driven by comparing fundamentally different kinds of mortgages.

3.7 Estimation details

The unit of observation is generally the bank-year, except in the final probit. Standard errors are clustered by bank. All regressions include year and state fixed effects.

Interpretation. Clustering by bank is important because the same bank appears repeatedly over time, and the paper's key variables vary a lot at the bank level.

4 Identification and instruments (what makes the estimates credible)

4.1 There is no formal IV

The paper does **not** use a separate instrumental variable. Instead, it uses an institutional discontinuity in loan liquidity created by the GSE charter rule and combines that with within-bank-year differencing.

Interpretation. Credibility comes from the design itself: the conforming cutoff creates a reason why two very similar mortgages can differ sharply in liquidity.

4.2 Why the jumbo cutoff is close to exogenous

The GSE cutoff is set by regulation rather than by any individual bank. A mortgage just below the cutoff can often be sold into the deep GSE-supported secondary market, while one just above the cutoff cannot.

Interpretation. This is the paper's key quasi-experimental source of variation. The cutoff changes the financing technology of the lender, not just the label of the loan.

4.3 Why differencing helps with demand

A simple regression of total originations on liquidity or deposit cost would be hard to interpret because strong loan demand can itself raise deposit rates or reduce holdings of liquid assets. By comparing nonjumbos to jumbos within the same bank-year, the paper attempts to difference out demand forces common to both segments.

Interpretation. This is the core identification logic. The bank faces the same local market, macro year, and broad borrower pool in both segments, but the two loan types differ in resale liquidity.

4.4 Why the filtered sample matters

The paper then narrows the sample by dropping refinancings and keeping only applications between 50% and 250% of the cutoff.

Interpretation. This reduces the chance that the estimated relative supply effect is really just a comparison between very small, atypical loans and very large, atypical loans.

4.5 Why acceptance rates strengthen the case

Using approval rates rather than raw originations adds another layer of control because the denominator is the application flow.

Interpretation. If funding conditions affect approvals, not just originations, that is more clearly a supply response of the lender itself.

4.6 Why sorting is both a threat and a result

Figure 1 makes clear that loan size is not fully exogenous near the cutoff. Borrowers may borrow less to stay under the cutoff, and banks with better funding conditions may attract borrowers seeking jumbo credit.

The paper treats this seriously rather than ignoring it:

- it studies application composition directly in Table V;
- it shows only about one-fifth of the full volume effect can be explained by approval-rate differences alone;
- it builds a separate homogeneous-sample test from loans that change status when the cutoff rises.

Interpretation. Sorting is not fatal to the paper. It is part of the mechanism through which lender financial condition affects observed credit allocation.

4.7 Placebo-style validation away from the cutoff

The paper asks whether the same financial effects show up at arbitrary size thresholds like 75% or 150% of the cutoff. They do not. The significant shift appears at the jumbo boundary.

Interpretation. This is very important. If liquidity effects shifted smoothly with loan size everywhere, the GSE-cutoff story would be less convincing. The fact that the action lines up with the institutional boundary supports the paper's interpretation.

4.8 Remaining limitations

The paper is careful, but several limitations remain:

- HMDA lacks rich borrower risk variables such as wealth, debt, loan-to-value, and property value at the loan level;
- lender-level mortgage pricing is not observed, so the paper cannot directly show how rates vary by lender around the cutoff;
- some nonjumbo loans are still not GSE-eligible for other reasons, which attenuates the liquidity difference and biases results toward zero;
- bank fixed effects reduce precision, especially for deposit cost, because most identifying variation is cross-sectional.

Interpretation. These limits do not overturn the main result, but they explain why the paper emphasizes quantities and approvals rather than direct price effects.

5 Tables and figures

5.1 Figure 1 and Table I: the cutoff matters in raw data

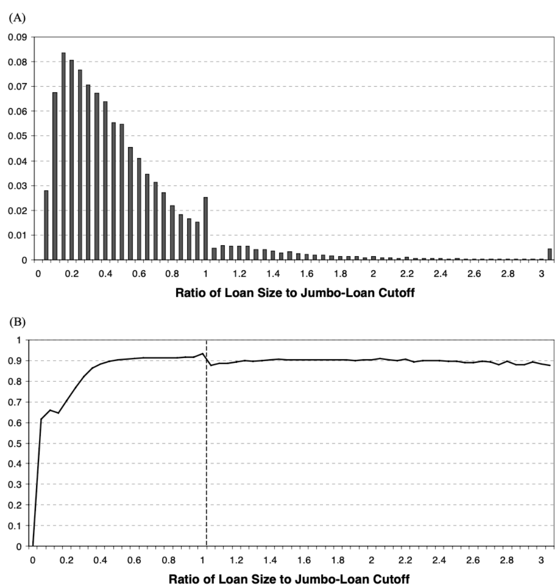

Figure 1. (A) Histogram of loan applications to all financial institutions, 1992–2004. This figure plots the frequency distribution of all mortgage applications in the HMDA data from 1992 to 2004 by the size of the application divided by the jumbo loan cutoff. For example, the vertical axis plots the share of mortgage applications greater than or equal to 95% of the jumbo loan cutoff but strictly less than the cutoff above the value 1 on the horizontal axis in the figure. **(B) Probability of acceptance for loan applications to all financial institutions, 1992–2004.** This figure plots the share of all mortgage applications that are accepted using all HMDA data from 1992 to 2004. We plot these acceptance rates by the size of the application divided by the jumbo loan cutoff. For example, the vertical axis plots the acceptance rate for mortgage applications that are greater than or equal to 95% of the jumbo loan cutoff but strictly less than the cutoff above the value 1 on the horizontal axis in the figure.

Table I

Summary Statistics for Mortgage Applications Characteristics

This table contains summary statistics for our two samples of mortgages. The first sample is based on all home mortgage applications to commercial banks from the HMDA data, collected by the Federal Reserve Board. The second sample includes only mortgage applications between 50% and 250% of the jumbo loan cutoff and omits refinancings.

	Full Sample	Mortgages between 50% and 250% of Jumbo Cutoff without Refis
Number of loan applications	62,587,880	6,435,270
Probability of acceptance (%)	79.93	90.85
Average loan amount (thousand)	$102	$209
Average applicant's income (thousand/year)	$79	$120
Average area income (thousand/year)	$44	$51
Average loan-to-income ratio	1.60	2.40
Percent of minority applicants	12.75	14.14
Percent of minority population in the area	16.81	14.83
Percent of female applicants	19.01	15.02
Percent of loans in MSA	84.17	91.84

Read: Figure 1 shows a clear bunching of applications just below the conforming threshold and a matching spike in approval rates just below the threshold. Table I then shows the two working samples. The full sample has about 62.6 million applications with an acceptance rate of roughly

79.9%. The filtered no-refi sample has about 6.4 million applications with an acceptance rate of about 90.9%, an average loan size of about \$209,000, and a loan-to-income ratio of about 2.40.

Economic message. The paper starts by documenting two things at once: a real liquidity discontinuity at the cutoff, and active borrower or lender behavior around that cutoff. That is why the later identification work must deal with both supply and sorting.

5.2 Table II: what the matched bank sample looks like

Read: The median matched bank has about \$84.4 million in assets, liquid assets equal to about 27.8% of assets, deposit cost of about 3.3%, and capital of about 8.5% of assets. The median bank originates nonjumbos equal to about 7.74% of assets and jumbos equal to about 1.38% of assets. Banks omitted from the matched sample are smaller and less mortgage-focused.

Economic message. The sample is made up of meaningful mortgage lenders, not tiny fringe institutions. This matters because the paper's bank-level financing variables are supposed to capture economically important constraints on loan supply.

Table II

Summary Statistics for Bank Characteristics

This table reports information on the distribution of characteristics for banks that we matched to the mortgage application data and for banks that we exclude from our analysis. Liquid assets equal cash plus marketable securities. The cost of deposits equals the ratio of interest expense on deposits to total deposits.

	Panel A: Banks Included in Our Sample			Panel B: Banks Excluded from Our Sample		
	25^{th} Percentile	Median	75^{th} Percentile	25^{th} Percentile	Median	75^{th} Percentile
Total assets (\$ million)	42.6	84.4	194.1	17.4	32.5	62.9
Financial structure and liquidity (%)						
Liquid assets/assets	19.9	27.8	37.4	22.9	32.7	43.9
Cost of deposits	2.7	3.3	3.9	2.7	3.5	4.1
Capital/assets	7.4	8.5	10.1	8.0	9.4	11.6
Loan shares (% of assets)						
Total loans	53.9	63.3	71.1	47.0	58.1	67.4
Commercial and industrial loans	9.7	17.6	27.4	14.4	25.7	41.4
Home mortgages	20.2	31.2	46.3	14.3	25.2	37.8
Commercial mortgages	10.7	18.1	26.9	4.9	10.7	19.4
Consumer loans	4.9	9.9	17.5	7.6	12.8	20.4
Loan originations (% of assets)						
Nonjumbo mortgages	3.09	7.74	16.26	N/A	N/A	N/A
Jumbo mortgages	0.65	1.38	3.05	N/A	N/A	N/A
Nonjumbo mortgages minus jumbos	1.33	5.57	13.41	N/A	N/A	N/A
Loan originations without refinancing (% of assets)						
Nonjumbo mortgages	1.56	3.83	7.55	N/A	N/A	N/A
Jumbo mortgages	0.44	0.91	1.97	N/A	N/A	N/A
Nonjumbo mortgages/less jumbo mortgages	0.44	2.46	5.87	N/A	N/A	N/A
Profit (%)						
Net income/assets	0.7	1.0	1.3	0.7	1.1	1.4
Mortgage applications						
Number of loan applications	90	240	627	N/A	N/A	N/A
Number of loans issued	79	215	564	N/A	N/A	N/A
Probability of acceptance (%)	85.5	91.5	95.6	N/A	N/A	N/A
Mortgage applications between 50% and 250% of jumbo cutoff without refis						
Number of loan applications	6	16	54	N/A	N/A	N/A
Number of loans issued	5	15	50	N/A	N/A	N/A
Probability of acceptance (%)	88.9	96.5	99.0	N/A	N/A	N/A

Table III

Regression of Loan Volumes for Nonjumbo Mortgages Relative to Jumbos on Bank Characteristics

This table reports regressions of the volume of approved nonjumbo minus jumbos mortgages, divided by beginning-of-period assets. The unit of observation is the bank year from 1992 to 2004. The regressions include the following controls for the loan pool characteristics in each bank year: the share of loans made to borrowers in MSAs; percent minority applicants in the bank's lending markets; mean loan-to-income ratio; log of mean applicant's income; average median income in bank's lending markets; the share of minority applicants; and the share of female applicants. We allow the coefficient on each of these variables to be different for the two segments of the market (jumbo and nonjumbo). Regressions also include three measures of local market demand conditions and demographics: the MSA-level unemployment rate (state for properties not in MSAs); the MSA-level growth in income (state for properties outside MSAs); and the percent of population over 65 for the state year. All regressions also include year and state fixed effects. *T*-statistics are in parentheses based on errors clustered at the bank level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	(Volume of Approved Nonjumbos − Volume of Jumbos)/Assets$_{t-1}$					
Dependent Variable	(1)	(2)	(3)	(4)	(5)	(6)
Bank's financial condition						
Liquid assets/assets	−0.166		−0.161	−0.099		−0.096
	(11.59)***		(11.31)***	(10.90)***		(10.71)***
Cost of deposits		1.001	0.704		0.657	0.489
		(4.51)***	(3.23)***		(4.79)***	(3.67)***
Bank controls						
Log of bank assets	−0.015	−0.015	−0.015	−0.009	−0.009	−0.009
	(10.49)***	(10.41)***	(10.49)***	(10.49)***	(10.41)***	(10.55)***
Bank owned by holding company	−0.021	−0.018	−0.019	−0.009	−0.007	−0.008
	(4.22)***	(3.56)***	(3.75)***	(3.09)***	(2.45)**	(2.54)**
Capital/assets	−0.224	−0.305	−0.207	−0.101	−0.147	−0.088
	(3.01)***	(4.09)***	(2.76)***	(2.46)**	(3.60)***	(2.15)**
Net income/assets	0.02	−0.138	0.005	−0.148	−0.222	−0.144
	(0.08)	(0.54)	(0.02)	(0.98)	(1.42)	(0.95)

(continued)

Table III—*Continued*

	(Volume of Approved Nonjumbos − Volume of Jumbos)/Assets$_{t-1}$					
Dependent Variable	(1)	(2)	(3)	(4)	(5)	(6)
Market controls						
Log of average application size[a]	−0.058	−0.057	−0.058	−0.038	−0.038	−0.038
	(5.63)***	(5.44)***	(5.62)***	(5.85)***	(5.70)***	(5.83)***
Percent over 65	0.046	0.045	0.048	0.013	0.013	0.014
	(4.07)***	(3.97)***	(4.23)***	(1.89)*	(1.82)*	(2.08)**
Local personal income growth	0.349	0.327	0.351	0.164	0.157	0.168
	(3.70)***	(3.48)***	(3.72)***	(2.93)***	(2.79)***	(2.99)***
Local unemployment rate	0.017	0.018	0.017	0.003	0.003	0.002
	(6.05)***	(6.30)***	(5.89)***	(1.67)*	(1.86)*	(1.45)
Loan pool characteristics included?	Yes					
Sample	All mortgages			Without refinancing		
Observations	32,982	32,982	32,982	27,372	27,372	27,372
R-squared	0.17	0.16	0.18	0.11	0.10	0.11

[a]The average applicant size reflects the typical size of mortgages in a lender's market. For banks lending in more than a single market (MSA), we construct an equally weighted average of the applicant sizes across markets.

5.3 Table III: volume regressions

Read: The dependent variable is the difference between approved nonjumbo and approved jumbo volumes, scaled by $Assets_{t-1}$. In the specification with both financial variables included, the liquidity coefficient is about −0.161 in the full sample and −0.096 in the sample without refinancings. The deposit-cost coefficient is about 0.704 in the full sample and 0.489 in the no-refi sample. The paper translates these into economically large shifts: moving liquidity from the 25th to the 75th percentile increases jumbo lending relative to nonjumbos by about 2.8% of assets in the full sample and 1.7% in the tighter sample. A similar move in deposit cost shifts lending toward nonjumbos by a bit more than 0.8% of assets in the full sample and about 0.5% in the tighter sample.

Economic message. This is the paper's central result. Better bank funding conditions matter a lot for illiquid mortgages and much less for liquid mortgages. The secondary market severs the usual link between bank finance and loan supply for nonjumbos.

5.4 Table IV: acceptance-rate regressions

Read: Now the dependent variable is $AR^{NJ} - AR^{J}$. In the specification with both financial variables, the liquidity coefficient is about −3.607 in the full sample and −2.142 in the tighter sample, while the deposit-cost coefficient is about 66.839 and 40.299, respectively. The signs match the theory: liquid banks and banks with cheaper deposits approve relatively more jumbo loans.

The paper's economic calibration implies that moving liquidity from the 25th to the 75th percentile raises the jumbo approval rate relative to the nonjumbo approval rate by about 0.6 percentage points.

Economic message. The same pattern survives when the outcome becomes a direct lending decision rather than raw origination volume. That makes the supply interpretation much more persuasive.

Table IV

Regressions of Acceptance Rates for Nonjumbo Mortgages Relative to Jumbos on Bank Characteristics

This table reports regressions of the acceptance rate for nonjumbo mortgages minus the acceptance rate for jumbos by bank year from 1992 to 2004. The regressions include the following controls for the loan pool characteristics in each bank year: the share of loans made to borrowers in MSAs; percent minority applicants in the bank's lending markets; loan-to-income ratio; log of mean applicant's income; average median income in bank's lending markets; share of minority applicants; and the share of female applicants. We allow the coefficient on each of these variables to be different for the two segments of the market—jumbo and nonjumbo. Regressions also include the log of application size, the MSA-level unemployment rate (state for properties not in MSAs); the MSA-level growth in income (state for properties outside MSAs); and the percent of population over 65 for the state year. All regressions also include year and state fixed effects. T-statistics are in parentheses based on errors clustered at the bank level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Acceptance Rate for Nonjumbos − Acceptance Rate for Jumbos					
Dependent Variable	(1)	(2)	(3)	(4)	(5)	(6)
Bank's financial condition						
Liquid assets/assets	−4.012		−3.607	−2.366		−2.142
	(4.28)***		(3.93)***	(2.61)***		(2.39)**
Cost of deposits		72.482	66.839		43.377	40.299
		(4.65)***	(4.35)***		(2.75)***	(2.57)**
Other bank controls						
Log of bank assets	−0.785	−0.79	−0.787	−0.32	−0.328	−0.324
	(7.60)***	(7.71)***	(7.67)***	(3.93)***	(4.04)***	(4.00)***
Bank owned by holding company	−1.134	−0.865	−0.866	−0.38	−0.214	−0.206
	(3.76)***	(2.84)***	(2.84)***	(1.25)	(0.68)	(0.66)
Capital/assets	4.77	4.769	6.581	0.432	0.154	1.248
	(1.37)	(1.38)	(1.86)*	(0.12)	(0.04)	(0.36)
Net income/assets	6.686	3.544	6.878	−14.589	−16.293	−14.664
	(0.44)	(0.23)	(0.45)	(0.91)	(1.01)	(0.91)
Loan pool characteristics included?	Yes					
Market controls included?	Yes					
Sample	All mortgages			Mortgages between 50% and 250% of jumbo cutoff without refis		
Observations	21,354	21,354	21,354	14,821	14,821	14,821
R-squared	0.10	0.10	0.10	0.03	0.03	0.03

Table V

Regressions of the Share of Nonjumbo Applications on Bank Characteristics

This table reports regressions of the share of applications that are nonjumbo by bank year from 1992 to 2004. The regressions include the following controls for the loan pool characteristics in each bank year: share of loans made to borrowers in MSAs; percent minority applicants in the bank's lending markets; loan-to-income ratio; log of applicant's income; average median income in bank's lending markets; share of minority applicants; and the share of female applicants. We allow the coefficient on each of these variables to be different for the two segments of the market–jumbo and nonjumbo. Regressions also include the log of application size in the MSA, the MSA-level unemployment rate (state for properties not in MSAs); the MSA-level growth in income (state for properties outside MSAs); and the percent of population over 65 for the state year. All regressions also include year and state fixed effects. T-statistics are in parentheses based on errors clustered at the bank level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Nonmarginal Applications: Number of Applications between 50% and 95% of Cutoff/Number of Applications between 50% and 95% Plus 105% and 250% of the Cutoff			Marginal Applications: Number of Applications between 95% and 100% of Cutoff/Number of Applications between 95% and 105% of the Cutoff		
Dependent Variable	(1)	(2)	(3)	(4)	(5)	(6)
Bank's financial condition						
Liquid assets/assets	−0.060		−0.061	−0.113		−0.112
	(4.26)***		(3.94)***	(6.25)***		(6.25)***
Cost of deposits		0.786	0.671		0.322	0.361
		(5.89)***	(5.26)***		(2.17)***	(2.01)**
Other bank controls						
Log of bank assets	0.007	0.007	0.007	0.017	0.016	0.017
	(5.83)***	(5.84)***	(5.83)***	(9.46)***	(9.45)***	(9.47)***
Bank owned by holding company	−0.007	−0.007	−0.007	0.015	0.015	0.016
	(2.26)**	(2.49)**	(2.48)**	(2.54)**	(2.57)**	(2.68)***
Capital/assets	−0.199	−0.198	−0.203	−0.294	−0.342	−0.289
	(4.53)***	(4.49)***	(4.58)***	(4.27)***	(4.97)***	(4.15)***
Net income/assets	0.487	0.498	0.492	−0.325	−0.405	−0.32
	(3.13)***	(3.21)***	(3.19)***	(1.23)	(1.51)	(1.21)
Loan pool characteristics included?	Yes					
Market controls included?	Yes					
Observations	34,306	34,306	34,306	11,056	11,056	11,056
R-squared	0.43	0.43	0.43	0.19	0.19	0.19

5.5 Table V: financially strong banks attract more jumbo applications

Read: Table V studies the share of applications that are nonjumbo. For nonmarginal applications, the liquidity coefficient is about −0.061 and the deposit-cost coefficient is about 0.671 in the specification with both variables. For marginal applications right around the cutoff, the corresponding coefficients are about −0.112 and 0.361. The paper summarizes the magnitude by noting that moving liquidity from the 25th to the 75th percentile raises the share of jumbo applications near the cutoff by about 1.8 percentage points.

Economic message. Approval decisions explain only part of the total lending effect. Borrowers with large desired loans also sort toward banks that are better positioned to fund illiquid mortgages, and some borrowers likely resize loans to stay below the cutoff.

5.6 Table VI: the effect shows up at the jumbo boundary, not at arbitrary size bins

Read: Table VI divides loans into four bins: 50–75%, 75–100%, 100–150%, and 150–250% of the cutoff. In both the volume and acceptance-rate specifications, the interactions for the 75–100% bin and the 150–250% bin are small and statistically weak. The important interaction is the jumbo-bin interaction. For example, in the acceptance-rate model with both financial variables, the liquidity × jumbo interaction is about 2.383 and the deposit-cost × jumbo interaction is about −71.859, both statistically strong.

Economic message. The key shift in sensitivity occurs where liquidity changes institutionally: at the conforming-loan boundary. That is strong evidence that the paper is capturing securitization-related liquidity rather than some generic loan-size effect.

Table VI

Regressions Analysis of Volumes and Acceptance Rates for Mortgages by Size Relative to the Jumbo Cutoff

This table reports regressions of the acceptance rate for mortgages of different sizes. For each bank year, there are four observations representing the acceptance rate for loans 50–75% of the jumbo cutoff, 75–100% of the cutoff, 100–150% of the cutoff, and 150–250% of the cutoff. The regressions include the following controls for the loan pool characteristics in each bank year: share of loans made to borrowers in MSAs; percent minority applicants in the bank's lending markets; loan-to-income ratio; log of applicant's income; average median income in bank's lending markets; share of minority applicants; and share of female applicants. The coefficients on each variable differ for the four loan size bins. These regressions also include indicators for each loan size bin as well as the other bank controls from the earlier tables (log of applicant size, log of bank's size, bank holding company indicator, capital/assets, and net income/assets). Regressions also include the following controls for demand and demographics: the MSA-level unemployment rate and income growth rate (state-level for properties outside MSAs) and the share of population over 65 in the state year. Regressions also include year and state fixed effects. T-statistics are in parentheses based on errors clustered at the bank level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Volume/Assets$_{t-1}$			Loan Acceptance Rate		
Dependent Variable	(1)	(2)	(3)	(4)	(5)	(6)
Bank's financial condition						
Liquid assets/assets	−0.222		−0.219	−0.521		−0.262
	(9.27)***		(9.11)***	(0.56)		(0.29)
Liquid assets/assets ∗ indicator for loans	0.044		0.042	1.298		1.115
(75–100% of jumbo cutoff)	(1.51)		(1.27)	(1.64)		(1.42)
Liquid assets/assets ∗ jumbo indicator	0.148		0.144	2.812		2.383
	(7.71)***		(7.52)***	(2.71)***		(2.32)**
Liquid assets/assets ∗ indicator for loans	0.01		0.007	0.61		0.571
(150–250% of jumbo cutoff)	(1.34)		(1.36)	(0.49)		(0.46)
Cost of deposits		0.802	0.476		45.633	46.049
		(1.29)	(1.36)		(1.53)	(1.57)
Cost of deposits ∗ indicator for loans		−0.191	−0.142		−14.781	−13.803
(75–100% of jumbo cutoff)		(1.36)	(1.02)		(1.54)	(1.47)
Cost of deposits ∗ jumbo indicator		−0.86	−0.727		−74.208	−71.859
		(6.31)***	(5.44)***		(5.59)***	(5.46)***
Cost of deposits ∗ indicator for loans		0.001	0.002		−9.279	−8.761
(150–250% of jumbo cutoff)		(0.01)	(0.03)		(0.57)	(0.54)
Other bank characteristics included?	Yes					
Loan pool characteristics included?	Yes					
Market controls included?	Yes					
Sample	All Mortgages between 50% and 250% of Jumbo Cutoff (Refis Included)					
Observations	131,854	131,854	131,854	106,192	106,192	106,192
R-squared	0.06	0.05	0.06	0.09	0.09	0.09

Table VII

Probit Regressions for Loan Acceptance Rates for Jumbo and Nonjumbo Mortgages Including Only Mortgages Near the Jumbo Cutoff

This table reports probit regressions of acceptance rates for mortgages of different sizes, where the unit of observation is the individual mortgage application. We include only mortgages that are nonjumbo but would be deemed jumbo during the preceding year, and mortgages that are jumbo but will be deemed nonjumbo in the subsequent year. From this pool of loans, we draw up to 1,000 applications for each bank year. In cases where there are fewer than 1,000 applications, we use all mortgage applications that meet the selection criterion defined above. We report marginal effects rather than probit coefficients. The probit regressions include the following additional controls: the jumbo loan indicator; the loan-to-income ratio; log of applicant's income; average median income in the property market; share of minority applicants; and share of female applicants. Regressions also include the following controls for demand and demographics: the MSA-level unemployment rate and income growth rate (state-level for properties outside MSAs) and the share of population over 65 in the state year. Regressions also include year and state fixed effects. T-statistics are in parentheses based on errors clustered at the bank level. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	1 if Loan Is Approved and 0 Otherwise					
Dependent Variable	(1)	(2)	(3)	(4)	(5)	(6)
Bank's financial condition						
Liquid assets/assets	0.026		0.025	0.004		0.003
	(0.96)		(0.96)	(0.18)		(0.14)
Liquid assets/assets ∗ jumbo indicator	0.056		0.061	0.021		0.018
	(3.39)***		(4.12)***	(2.48)***		(2.93)***
Cost of deposits		−0.039	0.012		−0.264	−0.261
		(0.11)	(0.04)		(0.72)	(0.72)
Cost of deposits ∗ jumbo indicator		−1.206	−1.248		−1.141	−1.133
		(5.05)***	(5.66)***		(4.94)***	(4.70)***
Other bank controls						
Log of bank assets	−0.011	−0.011	−0.011	−0.003	−0.003	−0.003
	(4.86)***	(5.21)***	(5.13)***	(1.78)*	(2.17)**	(2.15)**
Bank owned by holding company	−0.31	−0.311	−0.319	−0.05	−0.058	−0.062
	(2.10)**	(2.13)**	(2.20)**	(0.63)	(0.73)	(0.78)
Capital/assets	−0.015	−0.004	−0.007	−0.031	−0.025	−0.025
	(0.37)	(0.11)	(0.19)	(0.92)	(0.81)	(0.80)
Net income/assets	0.156	0.103	0.162	0.377	0.386	0.392
	(0.43)	(0.27)	(0.46)	(1.41)	(1.49)	(1.52)
Applicant characteristics included?	Yes					
Market controls included?	Yes					
Sample	Sampling from All mortgages			Sampling without Refis		
Observations	353,181	353,181	353,181	166,875	166,875	166,875
Pseudo R-squared	0.14	0.14	0.14	0.11	0.11	0.11

5.7 Table VII: near-cutoff homogeneous-sample probit

Read: Table VII keeps only loans that change classification in adjacent years when the jumbo cutoff rises. In the specification with both financial variables and controls, the marginal effect of liquidity on nonjumbos is essentially zero, but the liquidity × jumbo interaction is positive and significant: about 0.061 in the all-loans sample and 0.018 in the no-refi sample. Deposit cost again has little effect on nonjumbos, but the deposit-cost × jumbo interaction is strongly negative: about −1.248 in the all-loans sample and −1.133 without refis.

Economic message. Even when the paper compares only very similar loans near the moving cutoff, the same pattern survives: bank finance matters for illiquid mortgages and not for liquid ones. This is the paper's cleanest robustness result.

6 Short summary

- The paper asks whether securitization weakens the effect of bank balance-sheet liquidity and deposit funding costs on credit supply.
- It studies mortgages because conforming nonjumbo loans are much more liquid than jumbo loans due to GSE purchases and securitization.
- The empirical design compares nonjumbo and jumbo lending within the same bank-year, so common demand conditions can be differenced out.
- Using HMDA merged to Call Reports from 1992 to 2004, the paper finds that banks with more liquid assets and cheaper deposits originate relatively more jumbo mortgages.
- The same pattern appears in acceptance rates, which is stronger evidence of a supply effect.
- Approval behavior explains only part of the volume result; financially stronger banks also receive relatively more jumbo applications, especially near the cutoff.
- The finance effect appears at the jumbo boundary, not at arbitrary points in the loan-size distribution.
- A homogeneous-sample probit based on yearly changes in the conforming limit confirms that liquidity and deposit costs matter for jumbo approvals but not for nonjumbo approvals.

- The broader implication is that securitization reduces the sensitivity of credit supply to bank-specific funding shocks and therefore weakens the traditional bank lending channel.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that securitization weakens the link between a bank's own financial condition and its willingness to supply credit. In the mortgage market, balance-sheet liquidity and deposit funding costs matter for illiquid jumbo loans, but they matter much less for liquid nonjumbo loans that can be sold into the secondary market.

7.2 Economic intuition

A traditional bank that must keep a loan on balance sheet needs both funding and liquidity to support that origination. Once the loan can be sold or securitized, however, the bank no longer needs to finance the asset for long. That is why deposit costs and liquid-asset holdings strongly affect jumbo supply but have little effect on nonjumbo approvals. Securitization gives banks and mortgage markets an alternative funding technology.

7.3 What the paper adds to the literature

The paper sharpens the bank-lending-channel idea by showing that the effect of bank finance depends on *loan liquidity*. It is not just "banks versus markets." It is whether the specific loan can migrate from a bank balance sheet into a liquid secondary market.

7.4 Final takeaway

The deepest message is simple: when loans become tradable, local bank funding constraints matter less for who gets credit. That is good news if one worries about local credit crunches and monetary transmission through weak banks. But it also means that credit creation becomes more tightly linked to securitization markets and capital-market liquidity, with all of the integration benefits and systemic risks that became especially visible in the mortgage crisis.